

MIRPUR UNIVERSITY OF SCIENCE & TECHNOLOGY

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Software Design & Construction Lab Task

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Drone Delivery Optimization in Lahore

1. Executive Summary

This project evaluates two local search algorithms, Hill Climbing (HC) and Simulated Annealing (SA), for optimizing drone delivery routes in a simplified graph of Lahore, Pakistan. The objective is to find a route from Lahore Station to Emporium Mall while considering battery consumption, payload weight, airport restrictions, and traffic delays.

Both algorithms selected the route Lahore Station → Gulberg → Liberty Market → Emporium Mall. The total distance is 18 km, and the calculated battery consumption for a 3 kg payload is 97.92%. At the assumed drone speed of 40 km/h, the estimated delivery time is 27 minutes.

Hill Climbing has a lower execution time than Simulated Annealing in the recorded results. Payload testing shows that the selected route is feasible for payloads of 2 kg and 3 kg but exceeds the battery limit for payloads of 4 kg and 5 kg. The project also demonstrates how a heuristic-based initial route can fail the battery constraint and how dynamic traffic changes can affect route evaluation.

2. Problem Description and Constraints

The project represents a simplified delivery network as a weighted graph. Locations are represented by nodes, and the distances between connected locations are represented by edges.

The starting location is Lahore Station, and the destination is Emporium Mall.

The main constraints and parameters are:

- **Starting location:** Lahore Station.
- **Destination:** Emporium Mall.
- **Maximum nominal battery range:** 25 km without payload.
- **Payload:** 3 kg for the main algorithm comparison; 2 kg, 3 kg, 4 kg, and 5 kg for payload testing.
- **Payload penalty:** Battery consumption increases by 12% of the nominal consumption factor per kilogram of payload.
- **Airport restriction:** Allama Iqbal Airport is excluded as a route node.
- **Traffic penalty:** A 20% delivery-time penalty applies when a route includes Data Darbar, Walled City, or Badshahi Mosque.
- **Assumed drone speed:** 40 km/h.

3. Task 1: Hill Climbing Algorithm

3.1 Introduction

Hill Climbing is a local search algorithm that repeatedly examines neighboring solutions and selects an improvement according to an evaluation function. In this project, the algorithm uses heuristic-guided and random starting routes, evaluates neighboring routes, and attempts to reduce route cost.

The implementation uses multiple starting routes to reduce the chance of becoming stuck at a local optimum. It also evaluates battery consumption and excludes the airport node.

3.2 Implementation

The Hill Climbing function performs the following steps:

1. Generates an initial route using a heuristic and additional random starting routes.
2. Generates neighboring routes by changing sections of the current route.
3. Evaluates each neighboring route using the cost function.
4. Selects the neighboring route with the lowest cost when it improves the current solution.
5. Repeats the process until no improvement is found.
6. Returns the best route and its performance measurements.

3.3 Results

The initial greedy heuristic route was:

Lahore Station → Gulberg → DHA → Emporium Mall

This route has a distance of 23 km and consumes approximately 125.1% of the battery for a 3 kg payload. Therefore, it is infeasible.

Hill Climbing selected the following route:

Lahore Station → Gulberg → Liberty Market → Emporium Mall

Distance calculation:

8 km + 3 km + 7 km = **18 km**

Battery consumption is calculated using:

Battery consumption (%) = Distance × (1 + 0.12 × Payload) ÷ Maximum range × 100

For a distance of 18 km and a payload of 3 kg:

(18 × (1 + 0.12 × 3) ÷ 25) × 100 = **97.92%**

The route is feasible under the modeled battery limit, and its estimated delivery time is 27 minutes.

```
--- Greedy heuristic start (3 kg) ---
route       : Lahore_Station -> Gulberg -> DHA -> Emporium_Mall
distance_km : 23
battery_pct : 125.1
time_min    : 34.5
feasible    : False

--- Hill Climbing (3 kg) ---
route       : Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall
distance_km : 18
battery_pct : 97.9
time_min    : 27.0
feasible    : True
runtime_ms  : 2.044
```

4. Task 2: Simulated Annealing Algorithm

4.1 Introduction

Simulated Annealing is a probabilistic optimization algorithm inspired by the cooling process of heated materials. Unlike Hill Climbing, it can sometimes accept a worse neighboring solution to explore a wider search space and potentially avoid local optima.

In this project, Simulated Annealing generates alternative routes and evaluates them using the same route cost function, battery model, and airport-node restriction.

4.2 Implementation

The algorithm follows these steps:

1. Generates a random initial route.
2. Selects a neighboring route.
3. Calculates the difference between the candidate route's cost and the current route's cost.

4. Accepts an improved route or probabilistically accepts a worse route based on the current temperature.
5. Reduces the temperature using a cooling rate.
6. Continues until the temperature reaches the stopping threshold.
7. Returns the best route found, together with its distance, battery consumption, delivery time, and runtime.

The implementation uses an initial temperature of 50, a cooling rate of 0.9, a minimum temperature of 0.01, and 15 iterations per temperature level.

4.3 Results

For a 3 kg payload, Simulated Annealing selected the same route as Hill Climbing:

Lahore Station → Gulberg → Liberty Market → Emporium Mall

The results are:

- Total distance: 18 km
- Battery consumption: 97.92%
- Estimated delivery time: 27 minutes
- Route feasibility: True

Both algorithms therefore produced the same best-cost route in the recorded comparison, although their search strategies and execution times differ.

```
--- Simulated Annealing (3 kg) ---
route       : Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall
distance_km : 18
battery_pct : 97.9
time_min    : 27.0
feasible    : True
runtime_ms  : 369.066
```

5. Battery and Traffic Analysis

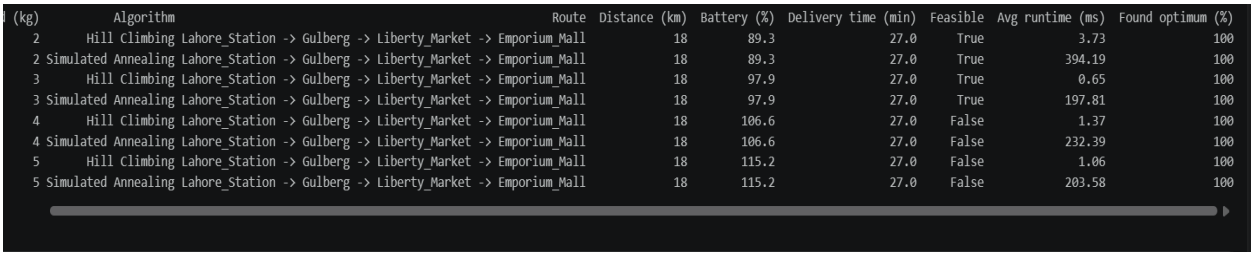
5.1 Battery Consumption Analysis

The battery model calculates consumption according to the route distance and payload weight. A heavier payload increases the estimated battery consumption.

For the selected 18 km route, the recorded results are:

Payload (kg)	Battery consumption	Feasible
2	89.3%	Yes
3	97.9%	Yes
4	106.6%	No
5	115.2%	No

The results show that the selected route satisfies the battery constraint for payloads of 2 kg and 3 kg. At 4 kg and 5 kg, battery consumption exceeds 100%, making the route infeasible under the implemented model.



l (kg)	Algorithm	Route	Distance (km)	Battery (%)	Delivery time (min)	Feasible	Avg runtime (ms)	Found optimum (%)
2	Hill Climbing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	89.3	27.0	True	3.73	100
2	Simulated Annealing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	89.3	27.0	True	394.19	100
3	Hill Climbing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	97.9	27.0	True	0.65	100
3	Simulated Annealing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	97.9	27.0	True	197.81	100
4	Hill Climbing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	106.6	27.0	False	1.37	100
4	Simulated Annealing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	106.6	27.0	False	232.39	100
5	Hill Climbing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	115.2	27.0	False	1.06	100
5	Simulated Annealing	Lahore_Station -> Gulberg -> Liberty_Market -> Emporium_Mall	18	115.2	27.0	False	203.58	100

5.2 Traffic Penalty Test

The code models traffic delays by applying a 20% penalty to the weighted route distance when a route includes Data Darbar, Walled City, or Badshahi Mosque.

For example, if the estimated delivery time before the penalty is 30 minutes, the adjusted time becomes:

30 × 1.20 = **36 minutes**

The selected 18 km route does not pass through these traffic-affected locations. Therefore, its estimated delivery time remains 27 minutes.

This test demonstrates the effect of the modeled traffic penalty. However, the implementation does not use real-time traffic data.

6. Results and Algorithm Comparison

The following table summarizes the recorded average runtimes for the 3 kg payload and the selected route.

Performance measure	Hill Climbing	Simulated Annealing

Selected route	Lahore Station → Gulberg → Liberty Market → Emporium Mall	Lahore Station → Gulberg → Liberty Market → Emporium Mall
Distance	18 km	18 km
Battery consumption	97.9%	97.9%
Estimated delivery time	27 minutes	27 minutes
Feasible	Yes	Yes
Average runtime	0.65 ms	197.81 ms
Best-cost result across recorded runs	100%	100%

The recorded comparison shows that Hill Climbing is considerably faster than Simulated Annealing for this problem. Both algorithms selected the same 18 km route and produced the same battery and delivery-time estimates.

The 100% result in the comparison represents how often the recorded runs matched the best cost observed in those runs. It should not be interpreted as independent proof of global optimality.

Execution times may vary between runs and computers.

7. Final Constraint Check

A constraint-checking function verifies the route's starting and ending locations, valid graph edges, airport-node exclusion, and battery consumption.

The results for the tested payloads are summarized below.

Payload	Correct start and destination	Valid edges	Airport node avoided	Battery within limit	Overall hard constraints
2 kg	Pass	Pass	Pass	Pass	Pass
3 kg	Pass	Pass	Pass	Pass	Pass
4 kg	Pass	Pass	Pass	Fail	Fail
5 kg	Pass	Pass	Pass	Fail	Fail

The selected route satisfies all checked hard constraints for 2 kg and 3 kg payloads. For 4 kg and 5 kg payloads, the route exceeds the battery limit and is marked infeasible.

The separate traffic indicator in the code reports whether a route passes through a traffic-affected location. The selected route does not pass through these locations.

Important limitation: Although the map displays a 5 km circle around Allama Iqbal Airport, the route-selection code excludes the airport node rather than checking every route segment against the full 5 km radius. Therefore, the implementation does not guarantee compliance with a true geographic no-fly perimeter.

Payload (kg)	Algorithm	Battery (%)	starts/ends correctly	valid edges only	avoids no-fly zone	battery <= 100%	passes Walled City (+20% time)	All hard constraints met
2	Hill Climbing	89.3	PASS	PASS	PASS	PASS	no	True
2	Simulated Annealing	89.3	PASS	PASS	PASS	PASS	no	True
3	Hill Climbing	97.9	PASS	PASS	PASS	PASS	no	True
3	Simulated Annealing	97.9	PASS	PASS	PASS	PASS	no	True
4	Hill Climbing	106.6	PASS	PASS	PASS	FAIL	no	False
4	Simulated Annealing	106.6	PASS	PASS	PASS	FAIL	no	False
5	Hill Climbing	115.2	PASS	PASS	PASS	FAIL	no	False
5	Simulated Annealing	115.2	PASS	PASS	PASS	FAIL	no	False

8. Dynamic Traffic Simulation

The project includes an additional simulation in which traffic conditions change during Simulated Annealing.

At step 300, the implementation assigns a very large cost multiplier to the Gulberg–Liberty Market edge to simulate severe congestion or closure. At step 700, the multiplier is cleared to represent traffic returning to normal.

The recorded output shows that the route being explored changes during the simulation. After traffic clears, the algorithm returns the original 18 km route in the recorded run.

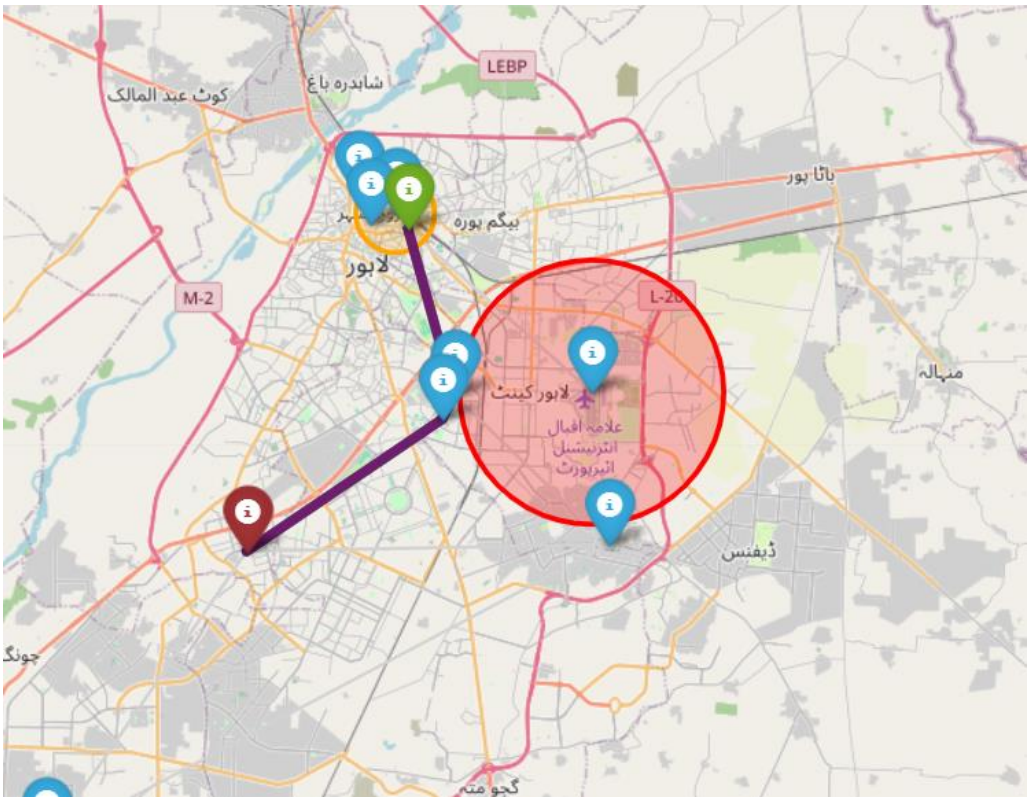
This experiment demonstrates that the route evaluation can respond to changing edge conditions. However, the traffic events are predefined in the code rather than obtained from real-time traffic services.

9. Route Visualization Using Folium

Folium is used to create an interactive map of the simplified Lahore delivery network. The map displays location markers, graph edges, the airport exclusion circle, the Walled City traffic zone, and the selected route.

The project also uses Matplotlib to generate static route diagrams. One figure compares the routes produced by Hill Climbing and Simulated Annealing for a 3 kg payload. Another illustrates why the initial greedy heuristic route is infeasible compared with the Hill Climbing result.

The map coordinates are approximate and illustrative. Geographic distances on the map may not match the distances defined in the graph.



10. Limitations

The project has several limitations:

1. The graph uses simplified distances rather than verified real-world flight distances.
2. The airport restriction excludes the airport node but does not enforce the complete 5 km radius against all route segments.
3. The traffic penalty is fixed and does not use real-time traffic information.
4. The battery model does not account for wind, weather, drone mass, battery degradation, or other physical operating factors.
5. Runtime measurements depend on the execution environment and can vary between runs.
6. Simulated Annealing uses randomized search, so its results can depend on the random seed and algorithm parameters.
7. The Folium map uses approximate coordinates, so it is intended for visualization rather than operational flight planning.

11. Future Improvements

The project could be improved by using accurate geographic coordinates and verified route distances. The airport restriction could be implemented by checking route segments against a properly defined no-fly boundary.

Real-time traffic and weather information could improve delivery-time estimates. The battery model could also include drone specifications, payload characteristics, weather effects, and a safety reserve.

Finally, more extensive experiments could compare repeated runs, different cooling schedules, and different initial solutions to evaluate algorithm performance more thoroughly.

12. Conclusion

This project demonstrates the use of Hill Climbing and Simulated Annealing for drone delivery route optimization on a simplified graph of Lahore.

Both algorithms selected the route Lahore Station → Gulberg → Liberty Market → Emporium Mall, with a total distance of 18 km. For a 3 kg payload, the calculated battery consumption is 97.92%, and the estimated delivery time is 27 minutes at the assumed speed of 40 km/h.

The recorded results show that Hill Climbing requires less execution time than Simulated Annealing for this problem. Payload testing demonstrates that the selected route is feasible for 2 kg and 3 kg payloads but exceeds the battery limit for 4 kg and 5 kg payloads.

The dynamic traffic experiment and route visualizations further demonstrate how local search algorithms can be applied to route planning. Although the model is simplified, it provides a practical example of optimization, constraint checking, and algorithm comparison in software development.